

MS101 EE Midsem

MS101 EE Quiz for first year UG students

1.

Which of the following unregulated DC power supplies has the highest peak-to-peak ripple voltage?

Marks: 1

Type: SINGLE_CORRECT_ANSWER

Options:

0)

An unregulated DC power supply made of a half-wave rectifier with load resistance $R_L = 2 \text{ k}\Omega$ and a filter capacitance $C = 100 \text{ }\mu\text{F}$.

1)

An unregulated DC power supply made of a half-wave rectifier with load resistance $R_L = 2 \text{ k}\Omega$ and a filter capacitance $C = 1000 \text{ }\mu\text{F}$.

2)

An unregulated DC power supply made of a bridge rectifier with load resistance $R_L = 2 \text{ k}\Omega$ and a filter capacitance $C = 100 \text{ }\mu\text{F}$.

3)

An unregulated DC power supply made of a bridge rectifier with load resistance $R_L = 2 \text{ k}\Omega$ and a filter capacitance $C = 1000 \text{ }\mu\text{F}$.

2.

A DC power supply is made using a step-down transformer (230 V rms to 15 V rms), and a half-wave rectifier circuit with a load resistance $R_L = 20 \text{ k}\Omega$ and a filter capacitance $C = 1000 \text{ }\mu\text{F}$.

You may assume the diode drop to be 0.6 V and the resistance of wires used for transformer windings to be negligible.

What is the peak output voltage of the DC power supply in volts?

Marks: 1.5

Type: FLOAT_TYPE

3.

Consider the binary number "10101011". This number is rotated right by 2 places. (For rotating right by one place, we move each bit one place to the right and bring the least significant bit of the original number to the most significant position). What is the rotated number in hexadecimal form?

4.

Consider the hexadecimal number “FA”. Write its value in decimal format if it is interpreted as:

4.1. A) an unsigned number

Marks: 1.5

Type: FLOAT_TYPE

4.2. B) a signed number

Marks: 1.5

Type: FLOAT_TYPE

5. Write the hexadecimal equivalent of the decimal number 325.

Marks: 1

Type: FILL_IN_THE_BLANKS_TYPE

6.

The truth table for a logical function F_n is given below. Using Karnaugh map (use the given Karnaugh map format) obtain the minimized logic expression F_n .

No.	A	B	C	D	F_n
00	0	0	0	0	1
01	0	0	0	1	1
02	0	0	1	0	1
03	0	0	1	1	1
04	0	1	0	0	0
05	0	1	0	1	1
06	0	1	1	0	0
07	0	1	1	1	0
08	1	0	0	0	1
09	1	0	0	1	0
10	1	0	1	0	0
11	1	0	1	1	0
12	1	1	0	0	0
13	1	1	0	1	1
14	1	1	1	0	0
15	1	1	1	1	0

AB↓ CD→	00	01	11	10
00				
01				
11				
10				

$F_n =$

Marks: 4

Type: IMAGE_ANSWER_TYPE

Rubrics:**Comments:**

7.

You are given a 4-bit number ABCD where A is the most significant bit. Obtain the logic expression F_n which will evaluate to TRUE only when the number represented by ABCD is exactly divisible either by 3 or by 7.

A) Write the truth table for the logical expression F_n .

B) Using Karnaugh map (use the given Karnaugh map format) obtain the minimized logic expression F_n .

AB↓ CD→	00	01	11	10
00				
01				
11				
10				

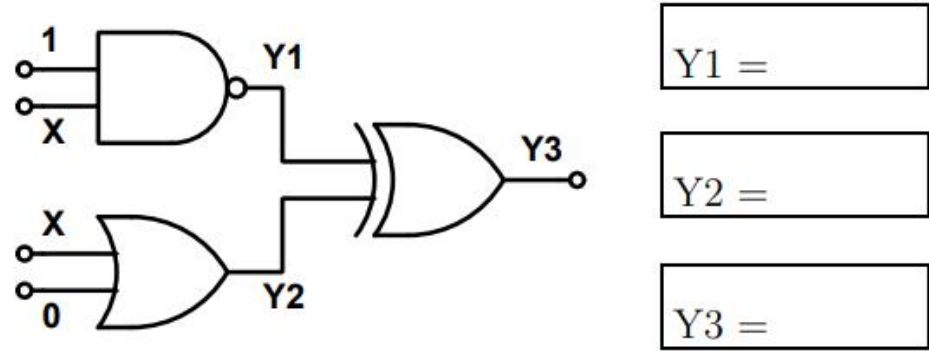
$F_n =$

Marks: 6

Type: IMAGE_ANSWER_TYPE

Rubrics:**Comments:**

8. Evaluate the outputs Y1, Y2 and Y3 in the following digital circuit.

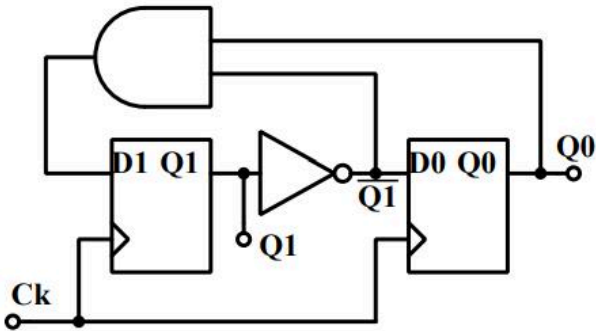


Rubrics:
Comments:

9.

Consider the digital circuit shown below. Initially, $Q1 = 1$ and $Q0 = 0$.

Write the decimal equivalent of the state $Q1 Q0$ after each clock (i.e. the 2-bit number $Q1 Q0$, where $Q1$ is the most significant bit) for the next 4 clocks.



Clock	Q1	Q0	Decimal Equivalent
0	1	0	2
1			
2			
3			
4			

9.1. Decimal equivalent of state $Q1 Q0$ after the 1st clock

9.2. Decimal equivalent of state $Q1 Q0$ after the 2nd clock

9.3. Decimal equivalent of state $Q1 Q0$ after the 3rd clock

9.4. Decimal equivalent of state $Q1 Q0$ after the 4th clock

10.

Arduino library provides the “map” function to map a value from one given range to another. It is called as:

$\text{newValue} = \text{map}(\text{oldValue}, \text{fromRangelow}, \text{fromRangehigh}, \text{toRangelow}, \text{toRangehigh})$.

What will be the returned value if the map function is called with the following arguments:

10.1. A) 512, 0, 1023, 0, 255

Marks: 2

Type: FLOAT_TYPE

10.2. B) 100, 0, 1023, 255, 0

Marks: 2

Type: FLOAT_TYPE

11.

The “analogWrite” function in Arduino library provides a pulse width modulated output on a specified pin. It divides 2ms in 255 time slots and if n is the PWM value given to the function, the output is at 5 V for n slots and at 0 V for the remaining $255-n$ slots. What value should we choose for n if we want the average of the output to be close to:

11.1. A) 1.0 V

Marks: 1.5

Type: FLOAT_TYPE

11.2. B) 2.9 V

Marks: 1.5

Type: FLOAT_TYPE

12.

Assume that an Analog-to-Digital Converter (ADC) is designed to report the digital value (represented value) corresponding to the nearest level which is less than the actual analog voltage. The input voltage of the ADC varies from 0 to 5 V. The input voltage at a given instant is 3.45 V. What will be the ADC output in the binary from if the ADC is:

12.1. A) a 1-bit ADC

Marks: 1

Type: FLOAT_TYPE

12.2. B) a 4-bit ADC

Marks: 2

Type: FLOAT_TYPE

13. Simplify the following Boolean expressions:

13.1. A) $X + 1 =$

Marks: 0.5

Type: FILL_IN_THE_BLANKS_TYPE

13.2. B) $X + X =$

Marks: 0.5

Type: FILL_IN_THE_BLANKS_TYPE

13.3. C) $X \cdot (X + Y) =$

Marks: 1

Type: FILL_IN_THE_BLANKS_TYPE

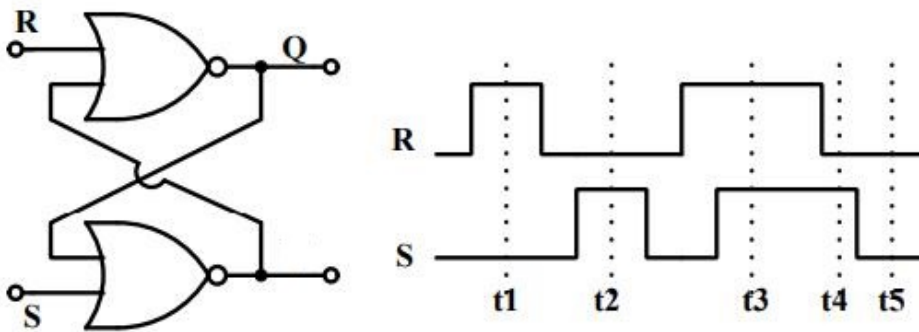
13.4. D) $X + \bar{X} \cdot Y =$

Marks: 1

Type: FILL_IN_THE_BLANKS_TYPE

14.

An R-S flip flop is initially in the Set state (i.e. $Q = 1$). The waveforms of the R and S inputs are as shown. Find the Q output values at the time instants t_1 , t_2 , t_3 , t_4 and t_5 .

14.1. A) Q output at time instant t_1

Marks: 0.5

Type: FLOAT_TYPE

14.2. B) Q output at time instant t_2

Marks: 0.5

Type: FLOAT_TYPE

14.3. C) Q output at time instant t_3

Marks: 0.5

Type: FLOAT_TYPE

14.4. D) Q output at time instant t_4

Marks: 0.5

Type: FLOAT_TYPE

14.5. E) Q output at time instant t_5

Marks: 0.5

Type: FLOAT_TYPE